

UNIVERSITY OF PETROLEUM AND ENERGY STUDIES

DEHRADUN, UK-IN



SCHOOL OF COMPUTER SCIENCE

OBJECT ORIENTED PROGRAMMING SYSTEM LAB (CSEG2120_1)

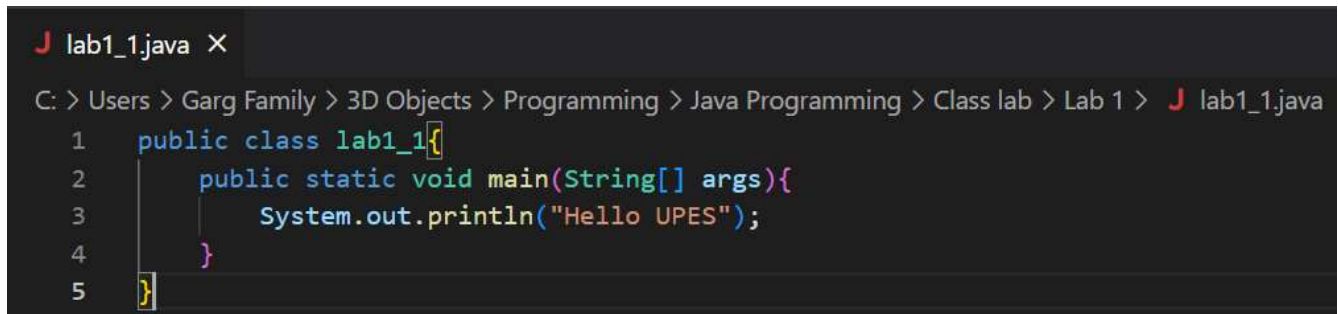
OOPS LAB FILE

Submitted To:
Dr. Suvojit Dhara
Assistant Professor
School of Computer Science

Submitted By:
Parth Garg
B. Tech CSE CSF
B-08 (4th Semester)
SAP ID :500122496

EXPERIMENT 1:

1. Write a simple Java Program to print “Hello UPES”.



```
lab1_1.java X
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 1 > lab1_1.java
1 public class lab1_1{
2     public static void main(String[] args){
3         System.out.println("Hello UPES");
4     }
5 }
```



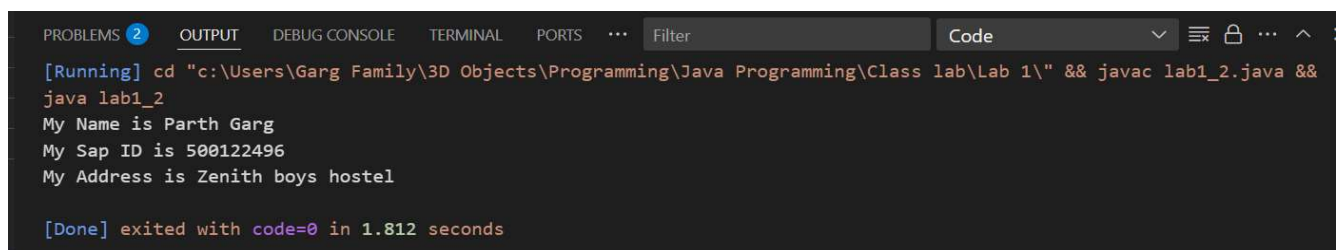
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS EXPOSED PORTS SPELL CHECKER Filter Code
[Running] cd "c:\Users\Garg Family\3D Objects\Programming\Java Programming\Class lab\Lab 1\" && javac lab1_1.java && java lab1_1
Hello UPES

[Done] exited with code=0 in 12.008 seconds
```

2. Write a simple Java Program that prints your name, SAP_ID, Batch, and address, all in different lines.



```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 1 > lab1_2.java >
You, 2 days ago | 1 author (You)
1 public class lab1_2 {
2
3     Run | Debug
4     public static void main(String[] args) {
5         System.out.println(x:"My Name is Parth Garg");
6         System.out.println(x:"My Sap ID is 500122496");
7         System.out.println(x:"My Address is Zenith boys hostel");
8     }
9 }
```



```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS ... Filter Code
[Running] cd "c:\Users\Garg Family\3D Objects\Programming\Java Programming\Class lab\Lab 1\" && javac lab1_2.java &&
java lab1_2
My Name is Parth Garg
My Sap ID is 500122496
My Address is Zenith boys hostel

[Done] exited with code=0 in 1.812 seconds
```

3. Design a visually appealing grade sheet that displays the Name, Roll Number, SAP ID, and Result. Use escape sequences and special characters like '*' to enhance its presentation. [No need to take any input from User].

C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 1 >  lab1_3.java

```
1
2 public class lab1_3 {
3
4     public static void main(String[] args) {
5         String name = "Parth Garg";
6         String rollNumber = "R2142231540";
7         String sapId = "500123456";
8         String result = "Pass";
9
10        System.out.println(x:"*****");
11        System.out.println(x:"*          GRADE SHEET          *");
12        System.out.println(x:"*****");
13        System.out.println("* Name      : " + name + "          *");
14        System.out.println("* Roll Number : " + rollNumber + "      *");
15        System.out.println("* SAP ID    : " + sapId + "            *");
16        System.out.println("* Result    : " + result + "          *");
17        System.out.println(x:"*****");
18    }
19 }
```

```
[Running] cd "c:\Users\Garg Family\3D Objects\Programming\Java Programming\Class lab\Lab 1\" && javac lab1_3.java && java lab1_3
```

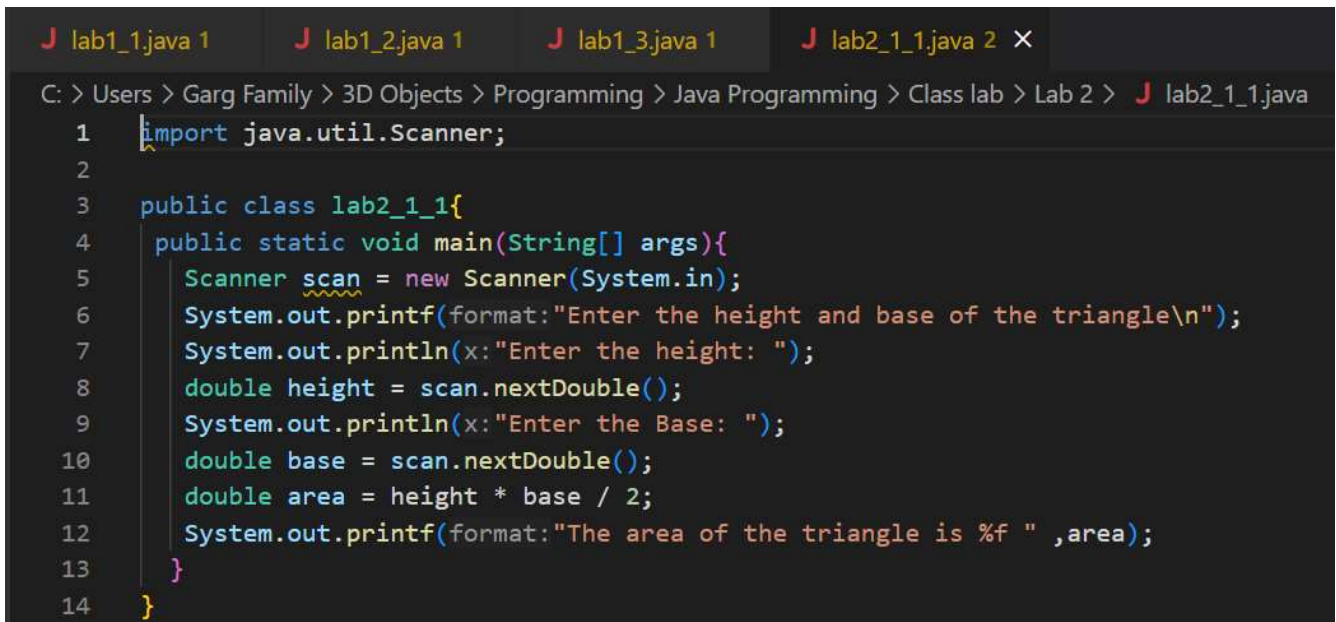
```
*****
*          GRADE SHEET          *
*****
* Name      : Parth Garg        *
* Roll Number : R2142231540     *
* SAP ID    : 500123456        *
* Result    : Pass             *
*****
```

```
[Done] exited with code=0 in 1.902 seconds
```

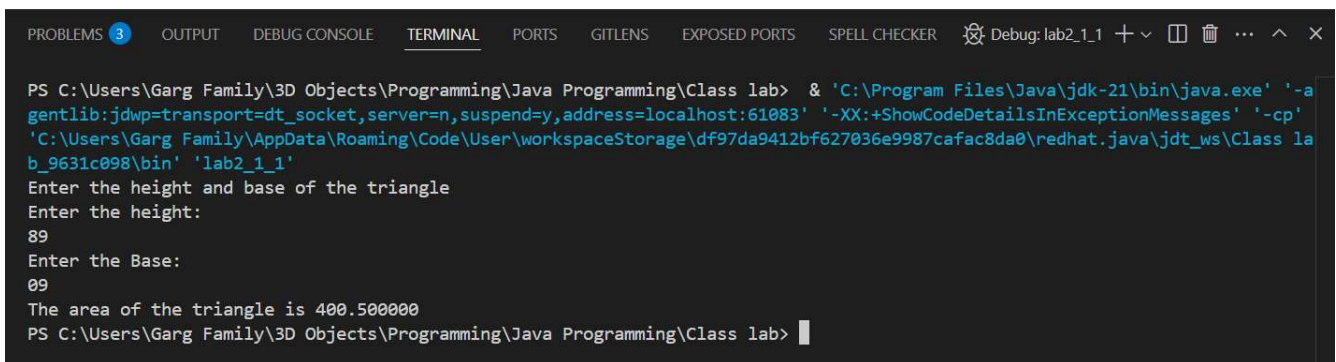
EXPERIMENT 2:

- 1) Write a program to find the area of a triangle.

1.1



```
lab1_1.java 1    lab1_2.java 1    lab1_3.java 1    lab2_1_1.java 2 X
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 2 > lab2_1_1.java
1  import java.util.Scanner;
2
3  public class lab2_1_1{
4      public static void main(String[] args){
5          Scanner scan = new Scanner(System.in);
6          System.out.printf(format:"Enter the height and base of the triangle\n");
7          System.out.println(x:"Enter the height: ");
8          double height = scan.nextDouble();
9          System.out.println(x:"Enter the Base: ");
10         double base = scan.nextDouble();
11         double area = height * base / 2;
12         System.out.printf(format:"The area of the triangle is %f " ,area);
13     }
14 }
```



```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS EXPOSED PORTS SPELL CHECKER Debug: lab2_1_1 + - [ ] ... ^ X
PS C:\Users\Garg Family\3D Objects\Programming\Java Programming\Class lab> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-a
gentlib:jdpw=transport=dt_socket,server=n,suspend=y,address=localhost:61083' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp'
'C:\Users\Garg Family\AppData\Roaming\Code\User\workspaceStorage\df97da9412bf627036e9987caf8da0\redhat.java\jdt_ws\Class la
b_9631c098\bin' 'lab2_1_1'
Enter the height and base of the triangle
Enter the height:
89
Enter the Base:
09
The area of the triangle is 400.500000
PS C:\Users\Garg Family\3D Objects\Programming\Java Programming\Class lab>
```

1.2

```

J lab1_1.java 1  J lab1_2.java 1  J lab1_3.java 1  J lab2_1_1.java 2  J lab2_1_2.java 2 X
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 2 > J lab2_1_2.java > ...
1  import java.util.*;          You, 3 months ago • Area of triangle ...
2
3  public class lab2_1_2{
4      public static void main(String[] args)
5      {
6          Scanner scan = new Scanner(System.in);
7          System.out.println(x:"Enter the sides of the triangle to calculate its area");
8          System.out.println(x:"Enter side 1:");
9          double S1 = scan.nextDouble();
10         System.out.println(x:"Enter side 2:");
11         double S2 = scan.nextDouble();
12         System.out.println(x:"Enter side 3:");
13         double S3 = scan.nextDouble();
14         double S = S1 + S2 + S3 / 2;
15         S = S*(S - S1)*(S - S2)*(S - S3);
16         double area = Math.sqrt(S);
17         System.out.println("The area of the triangle with sides " + S1 + S2 + S3 + " is " + area);
18     }
19 }

```

```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS EXPOSED PORTS SPELL CHECKER

PS C:\Users\Garg Family\3D Objects\Programming\Java Programming\Class lab> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:61112' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Garg Family\AppData\Roaming\Code\User\workspaceStorage\df97da9412bf627036e9987caf8da0\redhat.java\jdt_ws\Class lab_9631c098\bin' 'lab2_1_2'
Enter the sides of the triangle to calculate its area
Enter side 1:
78
Enter side 2:
4
Enter side 3:
34
The area of the triangle with sides 78.04.034.0 is 3582.9910689255144
PS C:\Users\Garg Family\3D Objects\Programming\Java Programming\Class lab> 

```

1.3

```

C: > Users > Garg Family > 3D Objects > Programming > Java Programming
1  public class lab2_1_3{          You, 3 months ago • Ar
2      public static void main(String[] args)
3      {
4          double b = Double.parseDouble(args[0]);
5          double h = Double.parseDouble(args[1]);
6          System.out.println("Area is " + b*h*0.5);
7      }
8  }

```



```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ javac lab2_1_3.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_1_3 10 90
Area is 450.0

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ |
```

2) Write a program to find simple interests.

2.1

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 2 > J lab2_2_1.java > ...
1  import java.util.*;
2
3  public class lab2_2_1{
4      public static void main(String[] args){
5          Scanner scan = new Scanner(System.in);
6          System.out.println(x:"This is a program to calculate the simple interest");
7          System.out.println(x:"Enter the principle: ");
8          double p = scan.nextDouble();
9          System.out.println(x:"Enter the rate: ");
10         double r = scan.nextDouble();
11         System.out.println(x:"Enter the time: ");
12         double t = scan.nextDouble();
13         double SI = p*r*t/100;
14         System.out.print("The simple interest is " + SI);
15     }
16 }
17 }
```

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS EXPOSED PORTS SPELL CHECKER Debug: lab2_2_1 + - [ ] ... ^ x
y,address=localhost:61468' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Garg Family\AppData\Local\Temp\vscodesws
_63a68\jdt_ws\jdt.ls-java-project\bin' 'lab2_2_1'
This is a program to calculate the simple interest
Enter the principle:
234500
Enter the rate:
2
Enter the time:
12
The simple interest is 56280.0
```

2.2

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming
1  public class lab2_2_2
2  {
3      public static void main(String[] args)
4      {
5          double p = Double.parseDouble(args[0]);
6          double r = Double.parseDouble(args[1]);
7          double t = Double.parseDouble(args[2]);
8          double SI = p*r*t/100;
9          System.out.println(SI);
10     }
11 }
12
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ javac lab2_2_2.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_2_2 10000 10 2
2000.0
```

- 3) Write a program to implement a command line calculator. (Try to Add sub Mul in the same program for 2 digits.)
(Hint: Integer.parseInt will be used)
For e.g. java calc 20 + 30
Output should be Sum of 20 and 30 is 50
java calc 50 * 30
Output should be Product of 50 and 30 is 1500

```

C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 2 > J lab2_3.java
1 public class lab2_3 {
2     public static void main(String[] args) {
3
4         // Parse command-line arguments
5         int num1 = Integer.parseInt(args[0]);
6         char symbol = args[1].charAt(index:0);
7         int num2 = Integer.parseInt(args[2]);
8
9         if (symbol == '+')
10            System.out.println(num1 + " + " + num2 + " is " + (num1+num2));
11        else if (symbol == '-')
12            System.out.println(num1 + " - " + num2 + " is " + (num1-num2));
13        else if (symbol == '*')
14            System.out.println(num1 + " * " + num2 + " is " + (num1*num2));
15        else
16            System.out.println(x:"Not a valid operation");
17    }
18 }

```

```

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ javac lab2_3.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_3 7+8
Exception in thread "main" java.lang.NumberFormatException: For input string: "7+8"
    at java.base/java.lang.NumberFormatException.forInputString(NumberFormatException.java:67)
    at java.base/java.lang.Integer.parseInt(Integer.java:661)
    at java.base/java.lang.Integer.parseInt(Integer.java:777)
    at lab2_3.main(lab2_3.java:5)

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_3 7 + 8
7 + 8 is 15

```

- 4) Write a Java program to check whether a given number is positive, negative, or zero using an if-else statement.

C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 2 >

```
1 import java.util.Scanner;
2
3 public class lab2_4{
4     public static void main(String[] args){
5         Scanner scan = new Scanner(System.in);
6         System.out.println(x:"Enter a number: ");
7         int num = scan.nextInt();
8         if (num > 0)
9             System.out.println(num + " is a positive number");
10        else if (num == 0)
11            System.out.println(num + " is zero");
12        else
13            System.out.println(num + " is a negative number");
14
15
16    }
17 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_4.java
Enter a number:
78
78 is a positive number
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_4.java
Enter a number:
0
0 is zero

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_4.java
Enter a number:
-9876
-9876 is a negative number
```

- 5) Create a program that accepts three integers and determines the greatest among them using nested if-else statements.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class  
  
1  import java.util.Scanner;  
2  
3  public class lab2_5{  
    Run main | Debug main | Run | Debug  
4      public static void main(String[] args){  
5          Scanner scan = new Scanner(System.in);  
6          System.out.println(x:"Enter 3 numbers: ");  
7          System.out.println(x:"Enter number1: ");  
8          int num1 = scan.nextInt();  
9          System.out.println(x:"Enter number2: ");  
10         int num2 = scan.nextInt();  
11         System.out.println(x:"Enter number3: ");  
12         int num3 = scan.nextInt();  
13  
14         if (num1 >= num2 && num1 >= num3)  
15             System.out.println(num1 + " is greatest");  
16         else if (num2 >= num1 && num2 >= num3)  
17             System.out.println(num2 + " is greatest");  
18         else  
19             System.out.println(num3 + " is greatest");  
20  
21     }  
22 }
```

You, 3 months ago • Nested if else

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)  
$ javac lab2_5.java  
  
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)  
$ java lab2_5.java  
Enter 3 numbers:  
Enter number1:  
67  
Enter number2:  
90  
Enter number3:  
4578  
4578 is greatest
```

- 6) Create a program that accepts a number (1–7) and displays the corresponding day of the week using a switch statement.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 2 > lab2_6.java
You, 3 months ago | 1 author (You)
1  import java.util.Scanner;
2
You, 3 months ago | 1 author (You)
3  public class lab2_6{
    Run main | Debug main | Run | Debug
4      public static void main(String[] args){
5          Scanner scan = new Scanner(System.in);
6          System.out.println(x:"Enter the number of the day of the week: ");
7          System.out.println(x:"Enter number: ");
8          int num = scan.nextInt();
9          switch(num)
10             {
11             case 1:
12                 System.out.println(x:"Monday");
13                 break;
14
15             case 2:
16                 System.out.println(x:"Tuesday");
17                 break;
18
19             case 3:
20                 System.out.println(x:"Wednesday");
21                 break;
22
23             case 4:
24                 System.out.println(x:"Thursday");
25                 break;
26
27             case 5:
28                 System.out.println(x:"Friday");
29                 break;
30
31             case 6:
32                 System.out.println(x:"Saturday");
33                 break;
34             }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_6
Enter the number of the day of the week:
Enter number:
6
Saturday
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_6
Enter the number of the day of the week:
Enter number:
45
```

Additional Question:

- 7) Write a program to calculate the final grade of a student based on the marks entered in three subjects. Use the following grading scale:

Average ≥ 90 : Grade A

Average ≥ 75 : Grade B

Average ≥ 50 : Grade C

Otherwise: Grade F

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab2_7
1  import java.util.Scanner;
2
3  public class lab2_7{
4      public static void main(String[] args){
5          Scanner scan = new Scanner(System.in);
6          System.out.println(x:"Enter Marks of 3 subjects: ");
7          System.out.println(x:"Enter Marks of Subject1: ");
8          int num1 = scan.nextInt();
9          System.out.println(x:"Enter Marks of Subject2: ");
10         int num2 = scan.nextInt();
11         System.out.println(x:"Enter Marks of Subject3: ");
12         int num3 = scan.nextInt();
13         int avg = num1 + num2 + num3 ;
14         avg = avg /3;
15
16         if (avg >= 90)
17             System.out.println(x:"Grade A");
18         else if (avg >= 75)
19             System.out.println(x:"Grade B");
20         else if (avg >= 50)
21             System.out.println(x:"Grade C");
22         else
23             System.out.println(x:"Grade F");
24
25     }
26 }
27 }
```

```
Garg Family@DESKTOP-QJNBVJU
$ java lab2_7
Enter Marks of 3 subjects:
Enter Marks of Subject1:
56 45 34
Enter Marks of Subject2:
Enter Marks of Subject3:
Grade F
```


- 8) Write a program to Take input as DD MM YYYY(04 08 2021) in the command line and calculate the number of days since 1 January 1970.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 2 > J lab2_8.java >
1  import java.time.LocalDate;
2  import java.time.temporal.ChronoUnit;
3
4  public class lab2_8 {
5      public static void main(String[] args) {
6
7          // Parse command-line arguments
8          int day = Integer.parseInt(args[0]);
9          int month = Integer.parseInt(args[1]);
10         int year = Integer.parseInt(args[2]);
11
12         // Define the given date and reference date (1 Jan 1970)
13         LocalDate givenDate = LocalDate.of(year, month, day);
14         LocalDate baseDate = LocalDate.of(1970, 1, 1);
15
16         // Calculate the number of days between the two dates
17         long daysSince1970 = ChronoUnit.DAYS.between(baseDate, givenDate);
18
19         // Print the result
20         System.out.println("Days since January 1, 1970: " + daysSince1970);
21     }
22 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ javac lab2_8.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_8
Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 0 out of bounds for length 0
    at lab2_8.main(lab2_8.java:8)

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_8 78 6 7989
Exception in thread "main" java.time.DateTimeException: Invalid value for DayOfMonth (valid values 1 - 28/31): 78
    at java.base/java.time.temporal.ValueRange.checkValidValue(ValueRange.java:319)
    at java.base/java.time.temporal.ChronoField.checkValidValue(ChronoField.java:718)
    at java.base/java.time.LocalDate.of(LocalDate.java:274)
    at lab2_8.main(lab2_8.java:13)

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_8 78 6 2025
Exception in thread "main" java.time.DateTimeException: Invalid value for DayOfMonth (valid values 1 - 28/31): 78
    at java.base/java.time.temporal.ValueRange.checkValidValue(ValueRange.java:319)
    at java.base/java.time.temporal.ChronoField.checkValidValue(ChronoField.java:718)
    at java.base/java.time.LocalDate.of(LocalDate.java:274)
    at lab2_8.main(lab2_8.java:13)

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 2 (main)
$ java lab2_8 8 6 2025
Days since January 1, 1970: 20247
```


EXPERIMENT 3:

- 1) Write a program to calculate the sum of all integers between 10 and 950 that are divisible by both 6 and 9.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_1.java > Language Support for Java(TM) by Red Hat > lab3_1
1 public class lab3_1{
2     public static void main(String[] args){
3         int sum = 0;
4         for(int i = 10 ; i <= 950 ; i++){
5             if(i % 6 == 0 && i % 9 == 0){
6                 sum += i;
7             }
8         }
9     }
10    System.out.println("The sum of all integers between 10 and 950 that are divisible by both 6 and 9 is " + sum);
11 }
12 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ javac lab3_1.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_1.java
The sum of all integers between 10 and 950 that are divisible by both 6 and 9 is 24804
```

- 2) 1(Extra). Find the GCD and LCM of two numbers m and n (user inputs).

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_1_2.java > Language Support for
You, 3 months ago | 1 author (You)
1 import java.util.Scanner;
2
You, 3 months ago | 1 author (You)
3 public class lab3_1_2{
4     Run main | Debug main | Run | Debug
5     public static void main(String[] args){
6         Scanner scan = new Scanner(System.in);
7         System.out.println(x:"Enter 2 numbers: ");
8         System.out.println(x:"Enter number1: ");
9         int num1 = scan.nextInt();
10        System.out.println(x:"Enter number2: ");
11        int num2 = scan.nextInt();
12        int num3 = num1 < num2? num1 : num2;
13        int product = num1*num2;
14        int hcf = 1;
15        for (int i = 1; i <= num3; i++) {
16            if (num1 % i == 0 && num2 % i == 0) {
17                hcf = i;
18            }
19        }
20        int lcm = product / hcf;
21
22        System.out.printf(format:"The HCF and LCM of %d and %d is %d and %d\n",num1,num2,hcf,lcm);
23    }
24
25 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_1_2.java
Enter 2 numbers:
Enter number1:
67
Enter number2:
9
The HCF and LCM of 67 and 9 is 1 and 603
```

- 3) Write a Java program that takes an integer as input and calculates the sum of its digits using a while loop.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_2
1  import java.util.Scanner;
2
3  public class lab3_2_1{
4      public static void main(String[] args) {
5          Scanner scan = new Scanner(System.in);
6          System.out.println(x:"Enter the number: ");
7          int num = scan.nextInt();
8          System.out.println("You entered: " + num);
9          int sum = 0 , a;
10         int b = num;
11         while (b > 0 ) {
12             a = b % 10;
13             sum = sum + a ;
14             b = b / 10 ;
15         }
16
17         System.out.println("The sum of digits of " + num + " is " + sum);
18     }
19 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_2_1.java
Enter the number:
67
You entered: 67
The sum of digits of 67 is 13
```

- 4) 2(Extra). Write a Java program that takes an integer as input and generates a number which has the digits in the reverse order.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_2_2.java >
You, 3 months ago | 1 author (You)
1  import java.util.Scanner;
2
3  public class lab3_2_2{
4      public static void main(String[] args) {
5          Scanner scan = new Scanner(System.in);
6          System.out.println(x:"Enter the number: ");
7          int num = scan.nextInt();
8          System.out.println("You entered: " + num);
9          int rev = 0 , a , b;
10         a = num;
11         while (a > 0 ) {
12             b = a % 10;
13             rev = b + rev * 10;
14             a = a / 10 ;
15         }
16
17         System.out.println("The reverse of " + num + " is " + rev);
18     }
19 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ javac lab3_2_2.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_2_2.java
Enter the number:
4567
You entered: 4567
The reverse of 4567 is 7654
```

5) Write a Java program that prints the first N terms of the Fibonacci series using a loop.

```
C:\> Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > J lab3_3_1.java > ...
You, 3 months ago | 1 author (You)
1  import java.util.Scanner;
2
3  public class lab3_3_1{
4      Run main | Debug main | Run | Debug
5      public static void main(String[] args) {
6          Scanner scan = new Scanner(System.in);
7          System.out.println(x:"How many terms of fibonacci series you want to print? ");
8          int n = scan.nextInt();
9          int a = 0, b = 1, c = 0;
10         System.out.println(x:"Fibonacci series: ");
11         for (int i = 0; i < n; i++) {
12             System.out.print(c + " ");
13             c = a + b;
14             a = b;
15             b = c;
16         }
17     }
18 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ javac lab3_3_1.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_3_1.java
How many terms of fibonacci series you want to print?
90
Fibonacci series:
0 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711 28657 46368 75025 121393 196418 317811 514229 832040 1346269 2178309 3524578 5702887 9227465
14930352 24157817 39088169 63245986 102334155 165580141 267914296 433494437 701408733 1134903170 1836311903 -1323752223 512559680 -811192543 -298632863 -1109825406 -1408458
269 1776683621 368225352 2144908973 -1781832971 363076002 -1418756969 -1055680967 1820529360 764848393 -1709589543 -944741150 1640636603 695895453 -1958435240 -1262539787 1
073992269 -188547518 885444751 696897233 1582341984 -2015728079 -433386095 1845853122 1412467027 -1036647147 375819880 -660827267 -285007387 -945834654 -1230842041 21182906
01 887448560 -1289228135 -401779575 -1691007710 -2092787285 511172301 -3581614984
```

- 6) 3(Extra). Write a Java program which checks whether a given number appears in the Fibonacci series or not.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_3_2.java > ...
You, 3 months ago | 1 author (You)
1  import java.util.Scanner;
2
3  public class lab3_3_2{
    Run main | Debug main | Run | Debug
4      public static void main(String[] args){
5          Scanner scan = new Scanner(System.in);
6          System.out.println(x:"Enter a number to check it comes in fibonacci series or not");
7          System.out.println(x:"Enter the number: ");
8          int num = scan.nextInt();
9
10         double n1 = 5*(num)*(num) + 4 , n2 = 5*(num)*(num) - 4;
11         double sq1 = Math.sqrt(n1), sq2 = Math.sqrt(n2);
12
13         if (sq1 == Math.floor(sq1) || sq2 == Math.floor(sq2)) {
14             System.out.println("Yes " + num + " comes in fibonacci series");
15         } else {
16             System.out.println("No " + num + " does not comes in fibonacci series");
17         }
18     }
19 }
20 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_3_2.java
Enter a number to check it comes in fibonacci series or not
Enter the number:
89007
No 89007 does not comes in fibonacci series

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ |
```

- 7) Write a Java program to count and display the total number of prime numbers between 1 and 1000.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_4_1.java > Language Support for Java(TM) by Red H
1  public class lab3_4_1
2  {
3      public static void main(String[] args)
4      {
5          int count = 0;
6          for(int i = 2 ; i <= 1000 ; i++)
7          {
8              if(isNumPrime(i))
9              {
10                 System.out.print(i + " ");
11                 count += 1;
12             }
13         }
14         System.out.printf(format:"\nThe total number of prime numbers found between 1 to 1000 is %d\n",count);
15     }
16
17     static boolean isNumPrime(int num)
18     {
19         if (num < 2) return false;
20         for (int i = 2; i <= Math.sqrt(num); i++)
21         {
22             if(num % i == 0)
23                 return false;
24         }
25         return true;
26     }
27 }
```



```

garg Family@DESKTOP-QJNBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_4_1.java
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179 181 191 193 197 199 211 223 227 229 2
33 239 241 251 257 263 269 271 277 281 283 293 307 311 313 317 331 337 347 349 353 359 367 373 379 383 389 397 401 409 419 421 431 433 439 443 449 457 461 463 467 479 487 4
91 499 503 509 521 523 541 547 557 563 569 571 577 587 593 599 601 607 613 617 619 631 641 643 647 653 659 661 673 677 683 691 701 709 719 727 733 739 743 751 757 761 769 7
73 787 797 809 811 821 823 827 829 839 853 857 859 863 877 881 883 887 907 911 919 929 937 941 947 953 967 971 977 983 991 997
The total number of prime numbers found between 1 to 1000 is 168

```

8) Write a Java program that counts how many times a given number appears in an array.

Input: arr = [2, 3, 2, 5, 2, 6], target = 2

Output: 3

```

J lab2_8.java 1 J lab3_1.java 1 J lab3_1_2.java 2 J lab3_2_1.java 2 J lab3_2_2.java 2 J lab3_3_1.jav
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > J lab3_5_1.java > Language Support
You, 3 months ago | 1 author (You)
1 import java.util.Scanner;
2
You, 3 months ago | 1 author (You)
3 public class lab3_5_1{
    Run main | Debug main | Run | Debug
4     public static void main(String[] args) {
5
6         Scanner scan = new Scanner(System.in);
7         System.out.println(x:"Enter the size of the Array");
8         int size = scan.nextInt();
9         int[] Array1 = new int[size];
10        System.out.println(x:"Enter the elements of thr Array: ");
11        for(int i = 0; i < size ; i++){
12            System.out.println("Enter element " + (i+1) + ": ");
13            Array1[i] = scan.nextInt();
14        }
15        System.out.println(x:"Entered Array is: ");
16        for(int x: Array1)
17            System.out.print(x + " ");
18
19        System.out.println();
20        System.out.println(x:"Enter the target you want to find in the Array: ");
21        int target = scan.nextInt();
22        int count = 0;
23        for(int x: Array1){
24            if(x == target)
25                count += 1;
26        }
27
28        if(count == 0)
29            System.out.println(target + " is not present in the given array");
30        else
31            System.out.println(target + " is present " + count + " times in the given array");
32    }
33 }

```

```

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_5_1.java
Enter the size of the Array
7
Enter the elements of thr Array:
Enter element 1:
90
Enter element 2:
4
Enter element 3:
3
Enter element 4:
2
Enter element 5:
67
Enter element 6:
4
Enter element 7:
0
Entered Array is:
90 4 3 2 67 4 0
Enter the target you want to find in the Array:
67
67 is present 1 times in the given array

```

- 9) 5(Extra) Write a Java program that takes an array from users and generates and displays another array of the unique elements from the former array.

```

C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_
You, 2 seconds ago | 1 author (You)
1
2 import java.util.Arrays;
3 import java.util.Scanner;
4
You, 2 seconds ago | 1 author (You)
5 public class lab3_5_2 {
6
Run main | Debug main | Run | Debug
7 public static void main(String[] args) {
8     Scanner scan = new Scanner(System.in);
9     System.out.println(x:"Enter the size of the Array: ");
10    int size = scan.nextInt();
11
12    int[] Array1 = new int[size];
13
14    System.out.println(x:"Enter the elements of the Array: ");
15    for (int i = 0; i < Array1.length; i++) {
16        Array1[i] = scan.nextInt();
17    }
18
19    int[] Array2 = new int[size];
20
21    int size2 = 0;
22    for (int i = 0; i < size; i++) {
23        // setting the flag to 0 every time
24        int flag = 0;
25
26        // checking if the element is present in the array or not
27        for (int y : Array2) {
28            if (y == Array1[i]) {
29                flag = 1;
30            }
31        }
32
33        //Passing the element in the array2 after checking
34        if (flag == 0) {

```

```
lab3_1.java 1    J lab3_1_2.java 2    J lab3_2_1.java 2    J lab3_2_2.java 2

C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab

5  public class lab3_5_2 {
7      public static void main(String[] args) {
28          //Passing the element in the array2 after checking
29          flag = 1;
30      }
31  }
32
33      //Passing the element in the array2 after checking
34      if (flag == 0) {
35          Array2[size2] = Array1[i];
36          size2 += 1;
37      }
38
39  }
40
41      System.out.println(x:"Entered Array is: ");
42      System.out.println(Arrays.toString(Array1));
43
44      System.out.println(x:"New Array is: ");
45      for (int i = 0; i < size2; i++) {
46          System.out.print(Array2[i] + " ");
47      }
48      System.out.println();
49  }
50 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_5_2.java
Enter the size of the Array:
3
Enter the elements of the Array:
1
1
1
Entered Array is:
[1, 1, 1]
New Array is:
1

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_5_2.java
Enter the size of the Array:
6
Enter the elements of the Array:
1
1
1
2
2
2
Entered Array is:
[1, 1, 1, 2, 2, 2]
New Array is:
1 2
```

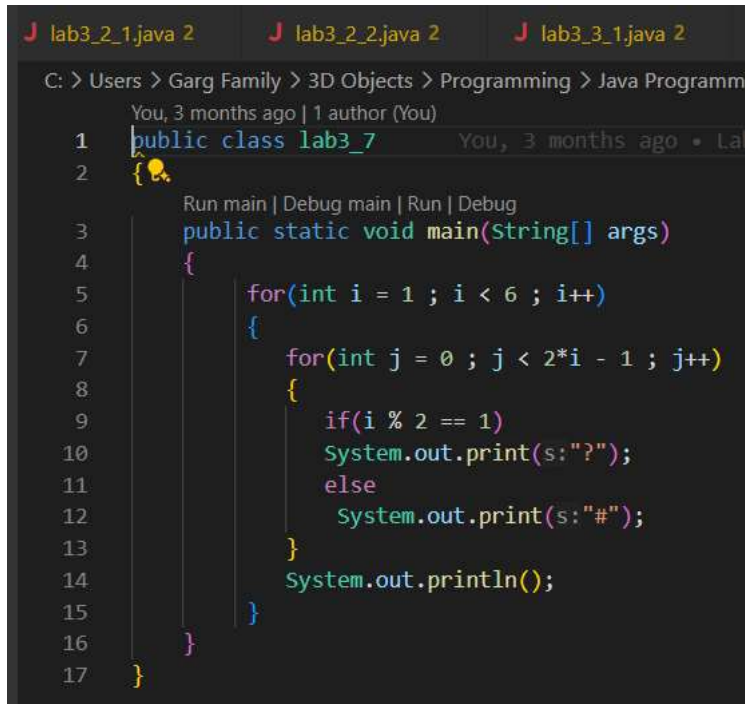
10) Write a Java program to find the second largest element in an integer array without sorting the array.

```
lab3_1_2.java 2 X  lab3_2_1.java 2  lab3_2_2.java 2  lab3_3_1.java 2  lab3_3_2,
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_6.
You, 3 months ago | 1 author (You)
1  import java.util.Scanner;
2  import java.util.Arrays;
3
You, 3 months ago | 1 author (You)
4  public class lab3_6
5  {
    Run main | Debug main | Run | Debug
6  public static void main(String[] args)
7  {
    You, 3 months ago * Lab work:
8      Scanner scan = new Scanner(System.in);
9      System.out.println(x:"Enter the size of the Array: ");
10     int size = scan.nextInt();
11
12     int[] Array1 = new int[size];
13
14     System.out.println(x:"Enter the elements of the Array: ");
15     for(int i = 0; i < Array1.length; i++)
16         Array1[i] = scan.nextInt();
17
18     int G1 = Array1[0];
19
20     for(int x: Array1)
21         G1 = G1 > x ? G1 : x;
22
23     int G2 = Array1[0];
24
25     for(int x: Array1)
26     {
27         if(x != G1)
28             G2 = G2 > x ? G2 : x;
29     }
30
31     System.out.println("Enter Array is: " + Arrays.toString(Array1));
32     System.out.println("Greatest Element is: " + G1);
33     System.out.println("Second greatest Element is: " + G2);
34 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_6.java
Enter the size of the Array:
6
Enter the elements of the Array:
11 22 33 44 55 66
Enter Array is: [11, 22, 33, 44, 55, 66]
Greatest Element is: 66
Second greatest Element is: 55
```


11) WAP to print the following pattern using loop

```
?  
###  
?????  
#####  
?????????
```



```
lab3_2_1.java 2    lab3_2_2.java 2    lab3_3_1.java 2  
C: > Users > Garg Family > 3D Objects > Programming > Java Programm  
You, 3 months ago | 1 author (You)  
1 public class lab3_7  
2 {  
    Run main | Debug main | Run | Debug  
3     public static void main(String[] args)  
4     {  
5         for(int i = 1 ; i < 6 ; i++)  
6         {  
7             for(int j = 0 ; j < 2*i - 1 ; j++)  
8             {  
9                 if(i % 2 == 1)  
10                    System.out.print(s:"?");  
11                else  
12                    System.out.print(s:"#");  
13            }  
14            System.out.println();  
15        }  
16    }  
17 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)  
$ java lab3_7.java  
?  
###  
?????  
#####  
?????????
```

12) Write a Java program that copies all elements from one array to another using a loop.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > J lab3_8.java > ...
You, 3 months ago | 1 author (You)
1 import java.util.Scanner;
2 import java.util.Arrays;
3
You, 3 months ago | 1 author (You)
4 public class lab3_8
5 {
    Run main | Debug main | Run | Debug
6     public static void main(String[] args)
7     {
8         Scanner scan = new Scanner(System.in);
9         System.out.println("Enter the size of the Array: ");
10        int size = scan.nextInt();
11
12        int[] Array1 = new int[size];
13
14        System.out.println("Enter the elements of the Array: ");
15        for(int i = 0; i < Array1.length; i++)
16            Array1[i] = scan.nextInt();
17
18        int[] Array2 = new int[size];
19
20        for(int i = 0; i < size; i++)
21            Array2[i] = Array1[i];
22
23        System.out.println("Entered Array is: ");
24        System.out.println(Arrays.toString(Array1));
25
26        System.out.println("New Array is: ");
27        System.out.println(Arrays.toString(Array2));
28
29    }
30 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_8.java
Enter the size of the Array:
5
Enter the elements of the Array:
7 8 5 9
Entered Array is:
[7, 7, 8, 5, 9]
New Array is:
[7, 7, 8, 5, 9]
```

13) Given an array containing N-1 unique numbers from 1 to N, write a Java program to find the missing number.

Input: [1, 5, 6, 2, 4]

Output: 3

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_9.java
You, 1 second ago | 1 author (You)
1
2 import java.util.Scanner;
3
4 public class lab3_9 {
5
6     public static void main(String[] args) {
7         Scanner scan = new Scanner(System.in);
8         System.out.println(x:"Enter the size of the Array: ");
9         int size = scan.nextInt();
10        int[] Array1 = new int[size - 1];
11
12        System.out.println(x:"Enter the elements of the array: ");
13        for (int i = 0; i < size; i++) {
14            System.out.println("Enter the element: " + (i + 1));
15            Array1[i] = scan.nextInt();
16        }
17        int sumOfN = size * (size + 1) / 2; // Sum of first N natural numbers
18        int sumOfArray = 0;
19
20        for (int i = 0; i < size - 1; i++) {
21            sumOfArray += Array1[i];
22        }
23
24        int missingNumber = sumOfN - sumOfArray;
25        System.out.println("The missing number is: " + missingNumber);
26    }
27 }
28
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ javac lab3_9.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_9.java
Enter the size of the Array:
5
Enter the elements of the array:
Enter the element: 1
1
Enter the element: 2
2
Enter the element: 3
4
Enter the element: 4
5
Enter the element: 5
6
The missing number is: 3
```

14) Write a Java program to rotate an array right by K positions.

Input: arr = [1, 2, 3, 4, 5], K = 2

Output: [4, 5, 1, 2, 3]

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 3 > lab3_10.java > ...
You, 3 months ago | 1 author (You)
1 import java.util.Scanner;
2
You, 3 months ago | 1 author (You)
3 public class lab3_10{
    Run main | Debug main | Run | Debug
4     public static void main(String[] args) {
5         Scanner scan = new Scanner(System.in);
6         System.out.println("Enter the size of the Array: ");
7         int size = scan.nextInt();
8         int[] Array1 = new int[size];
9
10        for(int i = 0 ; i < Array1.length; i++){
11            System.out.println("Enter the element " + (i+1) + ": ");
12            Array1[i] = scan.nextInt();
13        }
14
15        System.out.println("By how many positions you want to rotate the array right?");
16        int rotate = scan.nextInt();
17
18        rotate = rotate % size;
19
20        int[] Array2 = new int[size];
21
22        for (int i = 0; i < Array1.length; i++)
23            Array2[(i + rotate) % size] = Array1[i];
24
25
26        for (var x: Array1)
27            System.out.print(x + " ");
28        System.out.println();
29
30
31        for (var x: Array2)
32            System.out.print(x + " ");
33        System.out.println();
34    }
```



```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 3 (main)
$ java lab3_10.java
Enter the size of the Array:
7
Enter the element 1:
89
Enter the element 2:
3
Enter the element 3:
4
Enter the element 4:
324
Enter the element 5:
24
Enter the element 6:
5
Enter the element 7:
3
By how many positions you want to rotate the array right?
3
89 3 4 324 24 5 3
24 5 3 89 3 4 324
```

EXPERIMENT 4:

1. Create a Student class with attributes for name and age. Implement a default constructor to assign default values and a parameterized constructor to initialize the attributes with user-defined values. Create objects using both constructors and display their details.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 5

You, 29 seconds ago | 1 author (You)
3 public class Exp1 {
4
5     Run main | Debug main | Run | Debug
6     public static void main(String[] args) {
7         // Using default constructor
8         Student student1 = new Student();
9         student1.displayDetails();
10
11        // Using parameterized constructor
12        Student student2 = new Student(name:"Alice", age:20);
13        student2.displayDetails();
14    }
15
You, 29 seconds ago | 1 author (You)
16 class Student {
17
18     String name;
19     int age;
20
21     // Default constructor
22     Student() {
23         this.name = "Default Name";
24         this.age = 18;
25     }
26
27     // Parameterized constructor
28     Student(String name, int age) {
29         this.name = name;
30         this.age = age;
31     }
32
33     // Method to display student details
34     void displayDetails() {
35         System.out.println("Name: " + name + ", Age: " + age);
36     }
}
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 4 (main)
$ javac lab4_1.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 4 (main)
$ java lab4_1
The of the Student is Ansh and the age of the student is 20
The of the Student is Parth and the age of the student is 19

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 4 (main)
$ |
```

2. Create a BankAccount class with a private variable balance to store the account balance. Implement a public method deposit(double amount) to add funds, a protected method withdraw(double amount) to deduct funds, and a default-access method checkBalance() to display the current balance. Create an object of the class and demonstrate which methods and variables can be accessed both inside and outside the class

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 4 > lab4_2.java > Java > Ba

1
  You, 3 months ago | 1 author (You)
2 public class lab4_2 {
3
4     Run main | Debug main | Run | Debug
5     public static void main(String[] args) {
6         BankAccount MyAccount = new BankAccount();
7         MyAccount.deposit(amount:1000);
8         MyAccount.withdraw(amount:500);
9         System.out.println(x:"Account Balance: ");
10        MyAccount.CHECK_BALANCE();
11        // MyAccount.checkBalance(); error because the method is private
12    }
13 }
14
  You, 3 months ago | 1 author (You)
15 class BankAccount {
16
17     private double balance;
18     You, 3 months ago • lab4_2 ...
19     BankAccount() {
20     }
21
22     public void deposit(double amount) {
23         balance += amount;
24     }
25
26     protected void withdraw(double amount) {
27         if (balance >= amount) {
28             balance -= amount;
29         } else {
30             System.out.println(x:"The amount entered is more then the current balance");
31         }
32     }
33 }
```

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 4 > J lab4_2.java > Java > B
144
You, 3 months ago | 1 author (You)
15 class BankAccount {
16
17     private double balance;
18
19     BankAccount() {
20     }
21
22     public void deposit(double amount) {
23         balance += amount;
24     }
25
26     protected void withdraw(double amount) {
27         if (balance >= amount) {
28             balance -= amount;
29         } else {
30             System.out.println(x:"The amount entered is more then the current balance");
31         }
32     }
33
34     private void checkBalance() {
35         System.out.println("The current bank balance is: " + balance);
36     }
37
38     void CHECK_BALANCE() {
39         checkBalance();
40     }
41
42 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 4 (main)
$ ls
BankAccount.class  Course.class  Student.class      lab4_1.class  lab4_2.class  lab4_3.class  lab4_4.class  lab4_5.class
Calculator.class   LAB4.docx     University_Student.class  lab4_1.java   lab4_2.java   lab4_3.java   lab4_4.java   lab4_5.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 4 (main)
$ java lab4_2.java
Account Balance:
The current bank balance is: 500.0
```

3. Create a Calculator class that contains a method add() to perform addition. Overload the add() method to handle different types and numbers of parameters, such as adding two integers, two doubles, and three integers. Create an object of the class and demonstrate all method variations.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 4 > J lab4_3.java >
1
  You, 1 second ago | 1 author (You)
2 public class lab4_3 {
3
  Run main | Debug main | Run | Debug
4 public static void main(String[] args) {
5
6     Calculator cal = new Calculator();
7     System.out.println(cal.add(num1:5, num2:7));
8     System.out.println(cal.add(num1:15.89, num2:7.33));
9     System.out.println(cal.add(num1:5, num2:7.33));
10    System.out.println(cal.add(num1:5, num2:7, num3:8));
11    System.out.println(cal.add(s1:"Parth ", s2:"Garg"));
12
13 }
14 }
15 | You, 3 months ago • lab4_3 ...
  You, 1 second ago | 1 author (You)
16 class Calculator {
17
18     int add(int num1, int num2) {
19         return num1 + num2;
20     }
21
22     int add(int num1, int num2, int num3) {
23         return num1 + num2 + num3;
24     }
25
26     double add(double num1, double num2) {
27         return num1 + num2;
28     }
29
30     String add(String s1, String s2) {
31         return s1 + s2;
32     }
33 }
34 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 4 (main)
$ java lab4_3.java
12
23.22
12.33
20
Parth Garg
```


4. Create a University class that has a static variable universityName and a non-static variable studentName. Include a static method to display the university name. Then, create multiple student objects to demonstrate how the static variable is shared among all instances, while the non-static variable holds unique values for each object.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 4 > lab4_4.java > Language Support
You, 2 seconds ago | 1 author (You)
1
2 public class lab4_4 {
3
4     Run main | Debug main | Run | Debug
5     public static void main(String[] args) {
6
7         University_Student S1 = new University_Student();
8         S1.Student_Name = "Parth";
9         System.out.println("Student Name: " + S1.Student_Name + " University Name is ");
10        University_Student.Print();
11
12        University_Student S2 = new University_Student();
13        S2.Student_Name = "Ansh";
14        System.out.println("Student Name: " + S2.Student_Name + " University Name is ");
15        University_Student.Print();
16
17        University_Student S3 = new University_Student();
18        S3.Student_Name = "Gaurangi";
19        System.out.println("Student Name: " + S3.Student_Name + " University Name is ");
20        University_Student.Print();
21
22        University_Student S4 = new University_Student();
23        S4.Student_Name = "Himanshi";
24        System.out.println("Student Name: " + S4.Student_Name + " University Name is ");
25        University_Student.Print();
26
27        University_Student S5 = new University_Student();
28        S5.Student_Name = "Vanshika";
29        System.out.println("Student Name: " + S5.Student_Name + " University Name is ");
30        University_Student.Print();
31    }
32 }
33
```

```
31 }
32 }
33
34 You, 25 seconds ago | 1 author (You)
35 class University_Student {
36
37     static String University_Name = "UPES";
38     String Student_Name;
39
40     static void Print() {
41         System.out.println("University Name: " + University_Name);
42     }
43 }
44
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 4 (main)
$ java lab4_4.java
Student Name: Parth University Name is
University Name: UPES
Student Name: Ansh University Name is
University Name: UPES
Student Name: Gaurangi University Name is
University Name: UPES
Student Name: Himanshi University Name is
University Name: UPES
Student Name: Vanshika University Name is
University Name: UPES
```

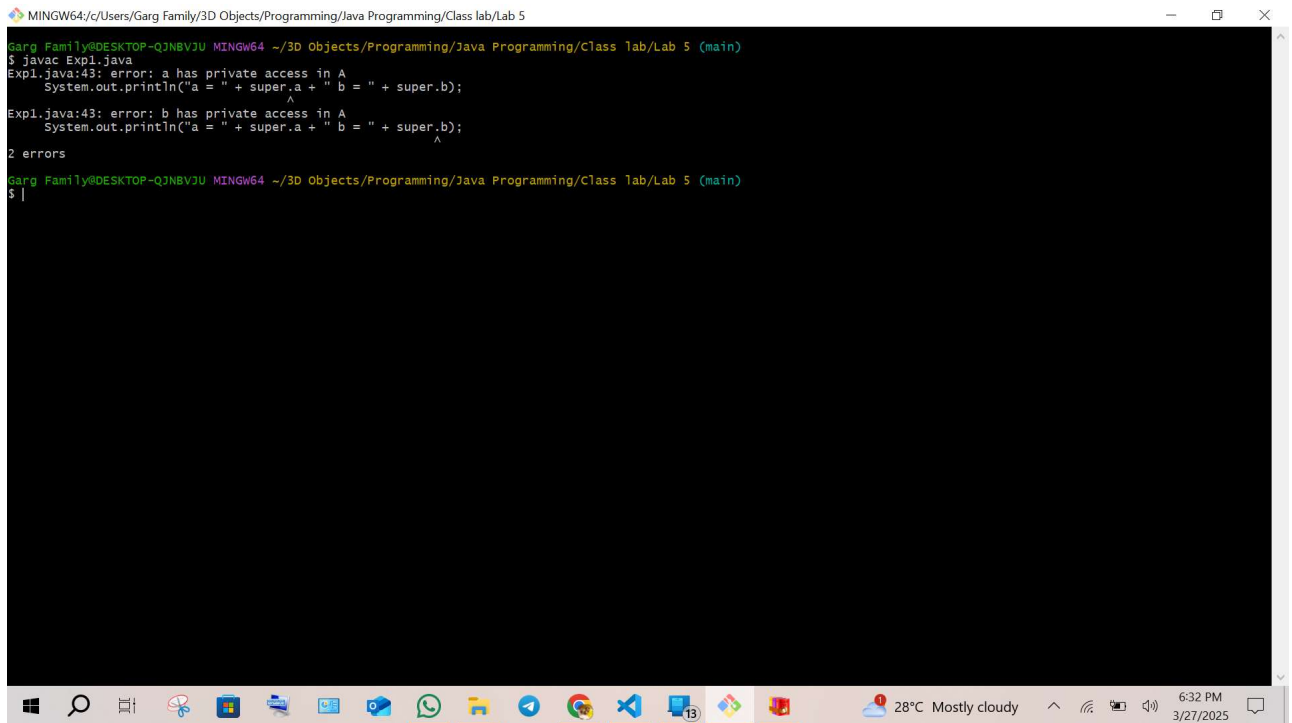
5. A student is developing a course registration system that allows students to enroll in courses. Each course has a course name and a course code. Implement a Course class with appropriate attributes and use the “this” keyword to differentiate between class attributes and constructor parameters during initialization. Create an object of the Course class and display the course details.

```
C: > Users > Garg Family > 3D Objects > Programming > Java Programming > Class lab > Lab 4 > lab4_5.java > ...
You, 3 months ago | 1 author (You)
1  You, 3 months ago • lab4 ...
You, 3 months ago | 1 author (You)
2  public class lab4_5 {
    Run main | Debug main | Run | Debug
3      public static void main(String[] args) {
4          Course course = new Course(courseName:"Introduction to Java Programming", courseCode:"1111");
5          course.displayCourseDetails();
6      }
7  }
8
You, 3 months ago | 1 author (You)
9  class Course {
10     String course_Name;
11     String course_Code;
12
13     Course(String courseName, String courseCode) {
14         this.course_Name = courseName;
15         this.course_Code = courseCode;
16     }
17
18     public void displayCourseDetails() {
19         System.out.println("Course Name: " + course_Name);
20         System.out.println("Course Code: " + course_Code);
21     }
22 }
```

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 4 (main)
$ java lab4_5.java
Course Name: Introduction to Java Programming
Course Code: 1111
```

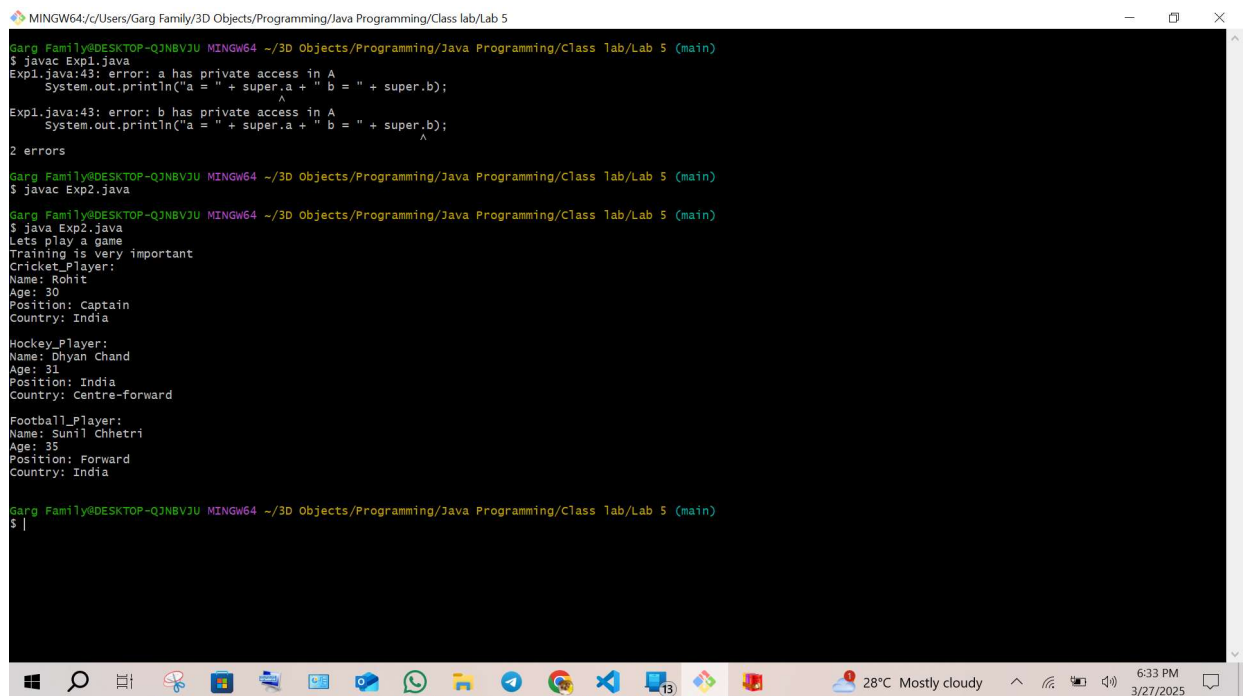
EXPERIMENT 5:

1. Write a Java program to demonstrate that a private member of a superclass cannot be accessed directly from a derived class.



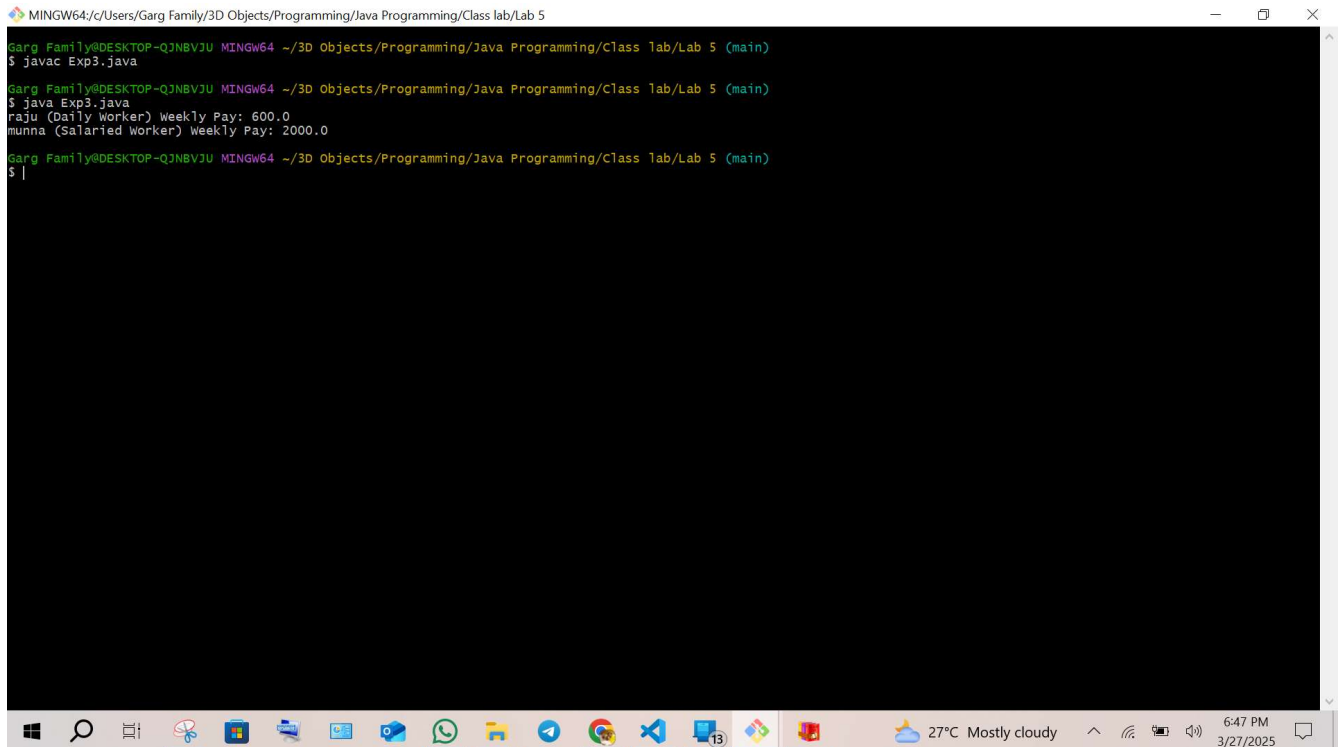
```
MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 5
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ javac Exp1.java
Exp1.java:43: error: a has private access in A
    System.out.println("a = " + super.a + " b = " + super.b);
                        ^
Exp1.java:43: error: b has private access in A
    System.out.println("a = " + super.a + " b = " + super.b);
                        ^
2 errors
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ |
```

2. Create a Java program with a Player class and derive three subclasses: Cricket_Player, Football_Player, and Hockey_Player. Implement attributes such as name, age, and position, and methods like play() and train() to represent these players.



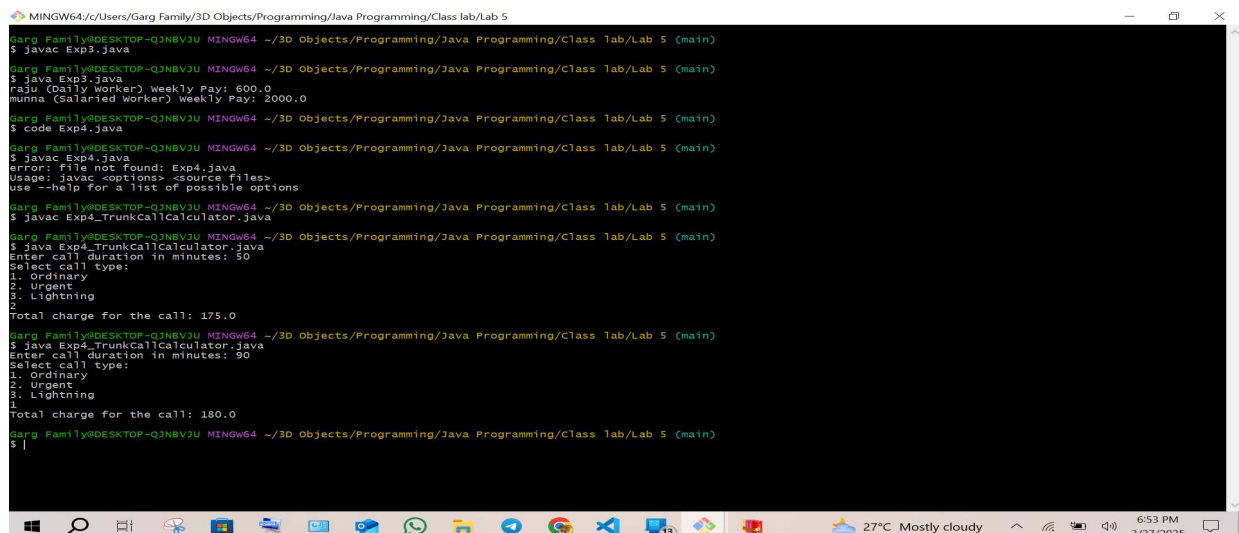
```
MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 5
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ javac Exp1.java
Exp1.java:43: error: a has private access in A
    System.out.println("a = " + super.a + " b = " + super.b);
                        ^
Exp1.java:43: error: b has private access in A
    System.out.println("a = " + super.a + " b = " + super.b);
                        ^
2 errors
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ javac Exp2.java
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ java Exp2.java
Lets play a game
Training is very important
Cricket_Player:
Name: Rohit
Age: 30
Position: Captain
Country: India
Hockey_Player:
Name: Dhyan Chand
Age: 31
Position: India
Country: Centre-Forward
Football_Player:
Name: Sunil Chhetri
Age: 35
Position: Forward
Country: India
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ |
```

3. Define a Worker class with DailyWorker and SalariedWorker as its subclasses. Each worker has a name and salary rate. Implement a method computePay(int hours) to compute weekly pay. DailyWorker is paid based on the number of days worked (assuming 8 hours per day), whereas SalariedWorker receives a fixed wage for 40 hours per week, regardless of actual hours worked. Use polymorphism to implement this program and test worker salary calculations.



```
MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 5
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ javac Exp3.java
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ java Exp3.java
raju (Daily worker) Weekly Pay: 600.0
munna (Salaried Worker) Weekly Pay: 2000.0
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ |
```

4. Implement a Java program to calculate trunk call charges based on duration and type (Ordinary, Urgent, or Lightning). Use polymorphism to manage different charge rates for each type. Implement a Java program to calculate trunk call charges based on duration (in minutes) and type (Ordinary, Urgent, or Lightning). Use polymorphism to manage different charge rates for each type. The program should take user input for duration and type and display the total charge.



```
MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 5
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ javac Exp3.java
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ java Exp3.java
raju (Daily worker) Weekly Pay: 600.0
munna (Salaried Worker) Weekly Pay: 2000.0
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ code Exp4.java
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ javac Exp4.java
error: file not found: Exp4.java
Usage: javac <options> <source files>
Use --help for a list of possible options
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ javac Exp4_TrunkCallCalculator.java
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ java Exp4_TrunkCallCalculator.java
Enter call duration in minutes: 50
Select call type:
1. Ordinary
2. Urgent
3. Lightning
2
Total charge for the call: 175.0
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ java Exp4_TrunkCallCalculator.java
Enter call duration in minutes: 50
Select call type:
1. Ordinary
2. Urgent
3. Lightning
2
Total charge for the call: 180.0
Garg Family@DESKTOP-Q3NBV3U MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ |
```


5. Design a Java class `Employee` with attributes `name`, `empid`, and `salary`. Implement a default constructor, a parameterized constructor, and methods to return the employee's name and salary. Add a method `increaseSalary(double percentage)` to raise the salary by a user-specified percentage. Create a subclass `Manager` with an additional instance variable `department`. Develop a test program to validate these functionalities.

```
MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 5
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ javac Exp5.java
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ java Exp5.java
Before Salary Increase:
Employee ID: 101
Name: Johnny lala
Salary: 50000.0

After 10% Salary Increase:
Employee ID: 101
Name: Johnny lala
Salary: 55000.0

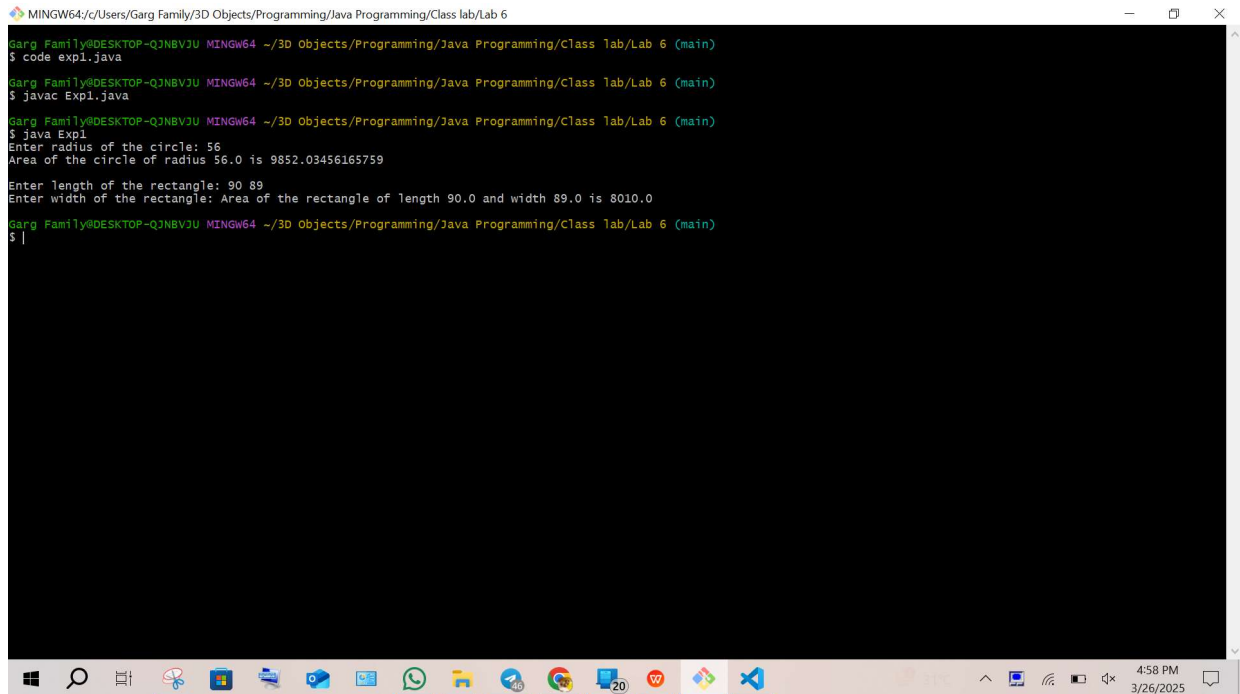
Manager Details:
Employee ID: 201
Name: chouhan shab
Salary: 70000.0
Department: IT

After 15% Salary Increase:
Employee ID: 201
Name: chouhan shab
Salary: 80500.0
Department: IT

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 5 (main)
$ |
```

EXPERIMENT 6:

1. Write a Java program to create an abstract class Shape with an abstract method calculateArea(). Derive two classes Rectangle and Circle from Shape and override the calculateArea() method to calculate and print the area of a rectangle and a circle, respectively. Use the main() method to create objects of Rectangle and Circle and test their calculateArea() methods.



```
MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 6
$ code expl.java
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ javac Expl.java
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ java Expl
Enter radius of the circle: 56
Area of the circle of radius 56.0 is 9852.03456165759
Enter length of the rectangle: 90
Enter width of the rectangle: 89
Area of the rectangle of length 90.0 and width 89.0 is 8010.0
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ |
```

2. Write a Java program to create an abstract class Employee with abstract methods calculateSalary() and displayDetails(). Derive two classes Manager and Developer from Employee and implement the methods to calculate the salary (e.g., based on fixed salary or hourly wage) and display employee details (e.g., name, role, salary). In the main() method, create objects of Manager and Developer and test their functionality.

```

MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 6
$ javac Exp2.java
error: Class names, 'Exp2.java', are only accepted if annotation processing is explicitly requested
1 error
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ javac Exp2.java
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ java Exp2
Name: Devi Singh
Age: 40
ID: 101
Position: Project Manager
Monthly salary: 156700.89
Yearly salary: 1880410.6800000002
Name: Rahul
Age: 24
ID: 102
Position: Software Developer
Enter hours worked since last payment:
Exception in thread "main" java.lang.NullPointerException: Cannot invoke "java.util.Scanner.nextInt()" because "this.scan" is null
    at Developer.calculateSalary(Exp2.java:84)
    at Exp2.main(Exp2.java:38)
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ javac Exp2.java
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ java Exp2
Name: Devi Singh
Age: 40
ID: 101
Salary:
Monthly salary: 156700.89
Yearly salary: 1880410.6800000002
Position: Software Developer
Name: Rahul
Age: 24
ID: 102
Salary:
Enter hours worked since last payment:
89
Your salary for 89 hours is $89000.0
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ |

```

3. Write a Java program to create an interface Bank with methods deposit(double amount) and withdraw(double amount). Implement this interface in a class Account that overrides these methods to perform deposit and withdrawal operations on a balance variable. Create another class BankDemo with a main() method to test the functionality by depositing and withdrawing amounts and displaying the updated balance.

```

MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 6
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ code Exp3.java
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ javac Exp3_BankDemo.java
Exp3_BankDemo.java:47: error: unreachable statement
    scan.close();
    ^
1 error
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ javac Exp3_BankDemo.java
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ java Exp3_BankDemo
Error: Could not find or load main class Exp3_BankDemo.
Caused by: java.lang.ClassNotFoundException: Exp3_BankDemo.
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ java Exp3_BankDemo
Your current balance is 0.0
Press 1 for deposit
Press 2 for withdrawal
Press 3 for checking current balance
Press 4 for exit
1
Enter the amount to deposit:
45
Press 1 for deposit
Press 2 for withdrawal
Press 3 for checking current balance
Press 4 for exit
3
Your current balance is 45.0
Press 1 for deposit
Press 2 for withdrawal
Press 3 for checking current balance
Press 4 for exit
4
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ |

```

4. Write a Java program to create an interface Playable with methods play(), pause(), and stop(). Implement this interface in a class MusicPlayer that overrides these methods to print appropriate messages (e.g., "Music is playing", "Music is paused", "Music is stopped"). Create another class TestPlayer with a main() method to test the functionality by calling the play(), pause(), and stop() methods.

```
MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 6
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ code Exp4.java
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ javac Exp4_TestPlayer.java
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ java Exp4_TestPlayer
Music is playing
Music is paused
Music is stopped
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 6 (main)
$ |
```

EXPERIMENT 7:

1. Create a package named Balance containing a class Account with a method Display_Balance() that displays the account balance. Write another program to import the Balance package and call the Display_Balance() method from the Account class.

```
MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 7/exp1
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7 (main)
$ cd exp1
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/exp1 (main)
$ ls
balance/ run.class run.java
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/exp1 (main)
$ java run
The balance is 5000000.0
Garg_Family@DESKTOP-Q3NBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/exp1 (main)
$ |
```


2. Create a package `p` containing a class `A` with four methods, each having different access modifiers (e.g., `public`, `protected`, `default`, and `private`). Write another class `B` in a different package `Q` with a `main()` method. Demonstrate how to access the methods of class `A` from class `B`, considering the access modifiers.

```
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/exp1 (main)
$ cd ..
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7 (main)
$ cd Exp2
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp2 (main)
$ ls
p/ q/
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp2 (main)
$ java q/B
Hi from Public method
Hallo from Protected method
Hola from default method
Hello from Private method
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp2 (main)
$ |
```

3. Write a Java program to demonstrate the use of the `final` keyword with a variable and a method. Create a class `MathConstants` with a `final` variable `PI` (value = 3.14159) and a `final` method `displayPI()` that prints the value of `PI`. Create another class `Circle` that extends `MathConstants` and includes a method `calculateArea(double radius)` to calculate and print the area of a circle using the formula: $\text{area} = \text{PI} * \text{radius} * \text{radius}$. In the `main()` method (in a separate class or `Circle`), test the `calculateArea()` method and observe the behavior when attempting to modify the `PI` variable or override the `displayPI()` method.

```
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp2 (main)
$ cd ..
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7 (main)
$ cd Exp3
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp3 (main)
$ ls
A.class Circle.class MathConstants.class 'final circle.PNG' 'final method1.PNG' 'final variable1.PNG' finalcode.class finalcode.java
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp3 (main)
$ java finalcode
The area of the circle of the radius 5.0 is 78.53975
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp3 (main)
$ |
```

4. Write a Java program to demonstrate the use of the `final` keyword with a class. Create a `final` class `Logger` with a method `logMessage(String message)` that prints the message to the console. Attempt to create another class `ExtendedLogger` that extends the `Logger` class and observe and explain the result. In the `main()` method (in a separate class), create an object of the `Logger` class and call the `logMessage()` method to print a sample message.

```
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp3 (main)
$ java finalcode
The area of the circle of the radius 5.0 is 78.53975
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp3 (main)
$ cd ..
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7 (main)
$ cd Exp4
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp4 (main)
$ ls
Logger.class 'final class child code.PNG' 'final class child error.PNG' finalclass.class finalclass.java
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp4 (main)
$ java finalclass
Hi Inside the final class
garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 7/Exp4 (main)
$ |
```

EXPERIMENT 8:

1. Create an outer class Library with a static nested class Book. The Book class should have attributes like title, author, and ISBN, and a method displayDetails() to print these details. In the main method, create an instance of the Book class and call displayDetails() to show the book information.

```
MINGW64/c/Users/Garg Family/3D Objects/Programming/Java Programming/Class lab/Lab 8

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ javac Expl.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Expl.java
Book Title: The Great Gatsby
Author: F. Scott Fitzgerald
ISBN: 9780743273565

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ javac Expl.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ javac Expl.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Expl.java
Title: Malgudi Days
Author: R. K. Narayan
-----
Title: The God of Small Things
Author: Arundhati Roy
-----
Title: Gitanjali
Author: Rabindranath Tagore
-----
Title: The Immortals of Meluha
Author: Amish Tripathi
-----
Title: The Namesake
Author: Jhumpa Lahiri
-----

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ |
```

2. Create an outer class Car with an inner class Engine. The Engine class should have a method start() that prints “Engine started” and a method stop() that prints “Engine stopped”. The Car class should have a method drive() that creates an instance of the Engine class and calls its start() and stop() methods.

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ javac Exp2.java
Exp2.java:24: error: cannot find symbol
    Car car = new Car();
                ^
  symbol:   class Car
  location: class Exp2
Exp2.java:24: error: cannot find symbol
    Car car = new Car();
                ^
  symbol:   class Car
  location: class Exp2
2 errors

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ javac Exp2.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Exp2.java
error: can't find main(String[]) method in class: Car

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ javac Exp2.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Exp2.java
error: can't find main(String[]) method in class: Car

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ AC

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Exp2
Engine started
Car is driving...
Engine stopped

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ |
```

3. Create an interface EventHandler with a method handleEvent(). In the main method, demonstrate the use of:

A local inner class inside a method registerEvent() that implements EventHandler and prints “Event handled by local inner class”.

An anonymous inner class that implements EventHandler and prints “Event handled by anonymous inner class”.

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ javac Exp3.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Exp3
Event handled by local inner class
Event handled by anonymous inner class

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ |
```

4. Write a Java program that takes two integers as input from the user and performs division. Handle the ArithmeticException that occurs if the denominator is zero. Use a try-catch block to catch the exception and display an appropriate error message. Additionally, use a finally block to print “Operation completed” regardless of whether an exception occurs or not.

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ javac Exp4.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Exp4
Enter numerator: 78
Enter denominator: 9
Result: 8
Operation completed.

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Exp4
Enter numerator: 78
Enter denominator: 0
Error: Cannot divide by zero.
Operation completed.

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ |
```

5. Write a Java program that creates an array of 5 integers and asks the user to enter an index to access the array element. Handle the `ArrayIndexOutOfBoundsException` if the user enters an invalid index. Use a try-catch block to catch the exception and display an appropriate error message. Use the finally block to print "Array access attempted".

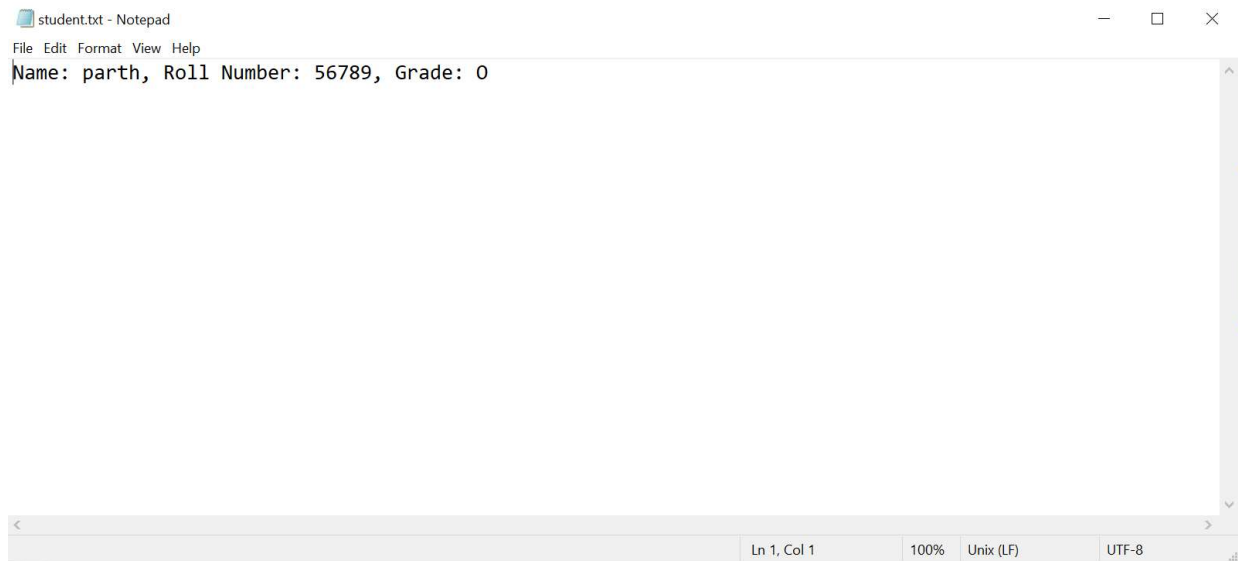
```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ javac Exp5.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Exp5
Enter an index (0 to 4): 4
Element at index 4 is: 50
Array access attempted.

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ java Exp5
Enter an index (0 to 4): 101
Error: Invalid index. Please enter a value between 0 and 4.
Array access attempted.

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 8 (main)
$ |
```

6. Write a Java program that takes user input for a student's name, roll number, and grade, and writes this information to a file named `student.txt` using `FileWriter`. Ensure the program appends the data to the file if it already exists. Handle any exceptions using try-catch and display an appropriate message if an error occurs.



Sample File Content:

Name: Aman, Roll Number: 120112, Grade: A

Name: Parul, Roll Number: 120131, Grade: B

7. Write a Java program that reads the contents of a file named student.txt using FileReader and displays the data on the console. Handle FileNotFoundException if the file does not exist and display an appropriate error message. Use a try-catch block for exception handling.

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class 1ab/Lab 8 (main)
$ javac Exp6.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class 1ab/Lab 8 (main)
$ java Exp6
Enter student's name: parth
Enter roll number: 56789
Enter grade: 0
Student information saved successfully.

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class 1ab/Lab 8 (main)
$ javac Exp7.java

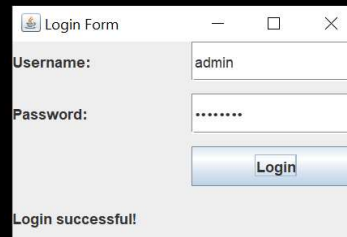
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class 1ab/Lab 8 (main)
$ java Exp7
Contents of student.txt:
Name: parth, Roll Number: 56789, Grade: 0

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class 1ab/Lab 8 (main)
$ |
```

EXPERIMENT 9:

- 1) Write a Java Swing program to create a Login Form using JTextField, JPasswordField, JButton, and JLabel. When the login button is clicked, validate if the username is “admin” and the password is “password”.

```
Garg_Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 9 (main)
$ javac Exp1.java
Garg_Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 9 (main)
$ java Exp1
```



- 2) Design a simple calculator using Java Swing with buttons for digits (0-9), addition (+), subtraction (-), multiplication (*), and division (/). Implement event handling for button clicks and display the result in a JTextField.

```
Garg_Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 9 (main)
$ javac Exp2.java
Garg_Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 9 (main)
$ java Exp2
```

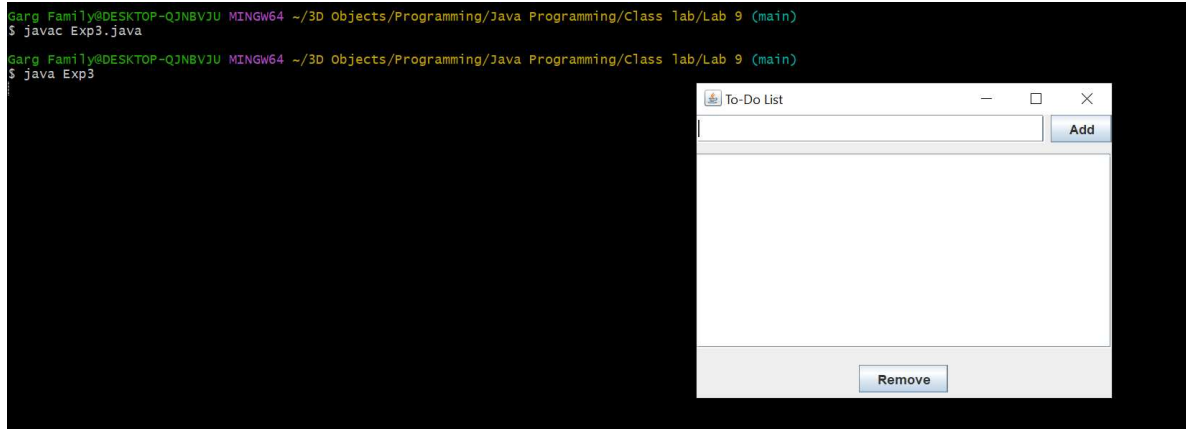


- 3) Write a Java Swing program to implement a To-Do List using JList. Provide::

A JTextField to enter tasks

An Add button to add tasks to the list

A Remove button to delete selected tasks

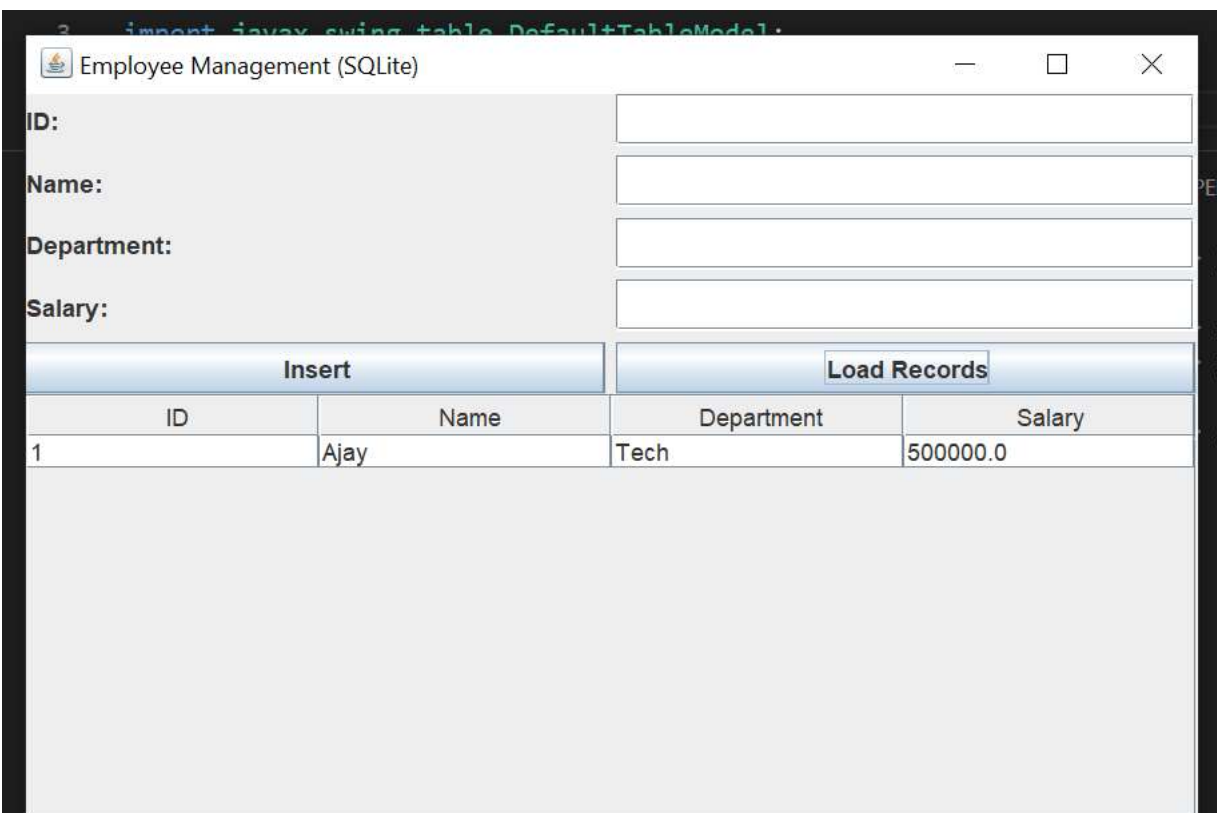


- 4) Write a Swing application with JDBC to insert and retrieve employee details (ID, Name, Department, Salary) into/from a MySQL database. Use

JTextField for input

JButton for inserting data

JTable for displaying retrieved records



EXPERIMENT 10:

1. Store a list of integers in an `ArrayList<Integer>` using autoboxing. Iterate through the list, unbox each value, and determine if it is a prime number, printing the results.

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10/src (main)
$ cd ..

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ javac -d bin src/Exp1.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ java -cp bin Exp1
Prime number check:
2 is a prime number.
3 is a prime number.
4 is not a prime number.
17 is a prime number.
18 is not a prime number.
19 is a prime number.
20 is not a prime number.

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ |
```

2. Create an `ArrayList` to store `Employee` objects with attributes like name, id, and salary. Add three employees, update one employee's salary, remove another by their id, and print the remaining employees.

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ javac -d bin src/Exp2.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ java -cp bin Exp2
Updated salary of ID 102 to ?58000
Removed employee with ID 101

Remaining employees:
ID: 102, Name: Parul, Salary: ?58000.0
ID: 103, Name: Ravi, Salary: ?60000.0

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ |
```

3. Use a `HashMap` to manage a product inventory where keys are `productId` (`Integer`) and values are `quantity` (`Integer`). Add three products, update the quantity of one product, remove another, and display the final inventory.

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ javac -d bin src/Exp3.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ java -cp bin Exp3
Updated quantity for product 102 to 45.
Removed product with ID 101.

Final Inventory:
Product ID: 102, Quantity: 45
Product ID: 103, Quantity: 20

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ |
```

4. Given an array of names with duplicates (e.g., ["Aman", "Varchasv", "Divyansh", "Varchasv", "Aman"]), store them in a HashSet to eliminate duplicates. Check if a specific name exists in the set and print the unique names.

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ javac -d bin src/Exp4.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ java -cp bin Exp4
Divyansh exists in the set.

Unique names:
Varchasv
Aman
Divyansh

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ |
```

5. Given an ArrayList of integers (possibly with duplicates), find the sum of its unique values. [Hint: Convert it to HashSet]

Example:

Input: [3, 5, 3, 8, 2, 5]

Unique values: [3, 5, 8, 2]

Sum: 18

```
Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ javac -d bin src/Exp5.java

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ java -cp bin Exp5
Unique values: [2, 3, 5, 8]
Sum of unique values: 18

Garg Family@DESKTOP-QJNBVJU MINGW64 ~/3D Objects/Programming/Java Programming/Class lab/Lab 10 (main)
$ |
```